ADVENTURES IN BIG MATH · 5TH GRADE MATH

MISSION 05 OF 08 · WORKSHEET PACK

Fraction Sums

Every graded set, every week — 5 weeks, 25 worksheets, with an answer key after each week.

WEEK 1	5 sets
WEEK 2	5 sets
WEEK 3	5 sets
WEEK 4	5 sets
WEEK 5	5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Equivalent Fractions

Same amount, cut more finely. Type fractions like 3/4.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up 6 ITEMS

1/2 = ?/4 — type the numerator	1/2 = ?/6	1/3 = ?/9
2/3 = ?/6	3/4 = ?/8	1/5 = ?/10

Core GRADED · SHOW YOUR WORKING

1. 2/5 = ?/15
2. 3/8 = ?/24
3. Simplify 6/8 — type as a/b
4. Simplify 10/15
5. Simplify 12/16

Challenge NOT GRADED

C1. Smallest common denominator of 1/4 and 1/6

C2. Smallest common denominator of 2/3 and 3/5

Adding Unlike Fractions

Convert both, then add numerators only. Answers here stay under two.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up
6 ITEMS

1/4 + 1/4 — type as a/b	1/3 + 1/3	1/8 + 3/8
2/5 + 1/5	1/6 + 1/6	3/10 + 2/10

Core
GRADED · SHOW YOUR WORKING

1. 1/2 + 1/4
2. 1/3 + 1/6
3. 2/3 + 1/6
4. 1/4 + 2/5
5. 3/8 + 1/4

Challenge
NOT GRADED

C1. 2/3 + 3/4

C2. 1/2 + 1/3 + 1/6

Subtracting Fractions

Same conversion. Borrowing turns one whole into a pile of pieces.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up

6 ITEMS

3/4 – 1/4 — type as a/b	5/6 – 1/6	7/8 – 3/8
1 – 1/2	1 – 1/4	4/5 – 2/5

Core

GRADED · SHOW YOUR WORKING

1. 1/2 – 1/3
2. 3/4 – 1/3
3. 1 – 3/8
4. 5/6 – 1/2
5. 2/3 – 1/4

Challenge

NOT GRADED

C1. 2 – 5/8 — type as a/b

C2. 1 1/4 – 2/3 — type as a/b

NAME __________

DATE __________

SCORE ____ / 11

Benchmarks

Is it nearer 0, a half, or 1? Estimate before you compute.

Warm-Up
6 ITEMS

Is 1/8 nearer 0 or 1 — type 0 or 1	Is 7/8 nearer 0 or 1	Is 4/8 exactly what — type as a/b
Is 5/6 nearer 1/2 or 1 — type 1/2 or 1	Is 2/5 nearer 0 or 1/2 — type 0 or 1/2	Larger: 3/4 or 1/2 — type it

Core
GRADED · SHOW YOUR WORKING

1. About how much is 7/8 + 1/8
2. Estimate 5/6 + 1/8 to the nearest half
3. Larger: 5/8 or 2/3 — type it
4. Larger: 3/5 or 5/8
5. Order 1/3, 1/2, 2/5 — type the smallest

Challenge
NOT GRADED

C1. Estimate 11/12 + 7/8 to the nearest whole

C2. Is 2/3 + 1/4 more or less than 1 — type more or less

Double the Recipe

Enrichment. Real amounts, doubled and halved.

NAME __________
DATE __________
SCORE ____ / 4

Warm-Up
2 ITEMS

Double 1/4 cup — type as a/b

Double 1/2 cup

Core
GRADED · SHOW YOUR WORKING

1. Half of 3/4 cup — type as a/b
2. Double 2/3 cup — type as a/b

Challenge
NOT GRADED

C1. A recipe needs 3/4 cup and you have 1/2 — how much more, as a/b

C2. Triple 3/8 cup — type as a/b

C3. 1/2 cup + 1/3 cup + 1/4 cup — type as a/b

TEACHER PAGE · ANSWER KEY

Week 1 Answers

Mission 05 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

1.1 · Equivalent Fractions

OUT OF 11

Warm-Up · 2 | 3 | 3 | 4 | 6 | 2

Core · 6 | 9 | 3/4 | 2/3 | 3/4

Challenge · 12 | 15

1.2 · Adding Unlike Fractions

OUT OF 11

Warm-Up · 1/2 | 2/3 | 1/2 | 3/5 | 1/3 | 1/2

Core · 3/4 | 1/2 | 5/6 | 13/20 | 5/8

Challenge · 17/12 | 1

1.3 · Subtracting Fractions

OUT OF 11

Warm-Up · 1/2 | 2/3 | 1/2 | 1/2 | 3/4 | 2/5

Core · 1/6 | 5/12 | 5/8 | 1/3 | 5/12

Challenge · 11/8 | 7/12

1.4 · Benchmarks

OUT OF 11

Warm-Up · 0 | 1 | 1/2 | 1 | 1/2 | 3/4

Core · 1 | 1 | 2/3 | 5/8 | 1/3

Challenge · 2 | less

Fri · Double the Recipe

OUT OF 4

Warm-Up · 1/2 | 1

Core · 3/8 | 4/3

Challenge · 1/4 | 9/8 | 13/12

NAME __________

DATE __________

SCORE ____ / 11

Subtract Unlike Fractions

Same conversion as adding, different sign. Type fractions like 3/4.

Warm-Up 6 ITEMS

3/4 – 1/4	5/6 – 1/6	7/8 – 3/8
4/5 – 2/5	2/3 – 1/3	9/10 – 4/10

Core GRADED · SHOW YOUR WORKING

1. 1/2 – 1/3
2. 3/4 – 1/3
3. 5/6 – 1/2
4. 2/3 – 1/4
5. 7/8 – 1/2

Challenge NOT GRADED

C1. 5/6 – 3/8

C2. 7/10 – 2/15

Borrowing from a Whole

1 becomes 5/5 so you have something to take from.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up

6 ITEMS

1 – 1/2	1 – 1/4	1 – 1/3
1 – 3/8	1 – 2/5	1 – 5/6

Core

GRADED · SHOW YOUR WORKING

1. 2 – 1/2
2. 2 – 3/4
3. 3 – 1/3
4. 2 – 5/8
5. 4 – 1/4

Challenge

NOT GRADED

C1. 2 – 5/6

C2. 3 – 7/8

NAME __________

DATE __________

SCORE ____ / 11

Mixed Numbers

Whole parts and fraction parts, handled separately.

Warm-Up

6 ITEMS

1 1/2 as an improper fraction	2 1/2 as an improper fraction	1 1/4 as an improper fraction
3/2 — the whole part	7/2 — the whole part	5/4 — the whole part

Core

GRADED · SHOW YOUR WORKING

1. 1 1/2 + 1/2
2. 2 1/4 + 1/4
3. 1 1/3 + 2/3
4. 2 3/4 – 1/4
5. 1 1/2 – 3/4

Challenge

NOT GRADED

C1. 2 1/3 + 1 1/2

C2. 3 1/4 – 1 1/2

NAME __________
DATE __________
SCORE ____ / 11

Fraction Word Problems

Two-step problems with a subtraction hiding inside.

Warm-Up
6 ITEMS

Ate 1/4, then 1/4 more — total	Had 3/4, ate 1/4 — left	1/2 hour plus 1/2 hour — hours
1 whole minus 1/3	1/8 + 1/8	2/5 + 1/5

Core
GRADED · SHOW YOUR WORKING

1. Walked 1/2 mile then 1/4 — total ....
2. A 3/4 cup recipe, you have 1/2 — how much more ....
3. Read 2/3 then 1/6 more — total ....
4. Ran 5/6 km, walked 1/3 km — total ....
5. 1 hour minus 1/4 hour — minutes ....

Challenge
NOT GRADED

C1. A tank 3/4 full, 1/3 used — fraction left

C2. 2/3 of an hour in minutes

NAME __________

DATE __________

SCORE ____ / 4

Double the Recipe Begins

Pick the recipe, write the scaled amounts.

Warm-Up

2 ITEMS

Double 1/4 cup

Double 1/2 cup

Core

GRADED · SHOW YOUR WORKING

1. Double 2/3 cup
2. Half of 3/4 cup

Challenge

NOT GRADED

C1. Triple 3/8 cup

C2. 1/2 + 1/3 + 1/4 cup — total

C3. Double 1 1/2 cups

TEACHER PAGE · ANSWER KEY

Week 2 Answers

Mission 05 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

2.1 · Subtract Unlike Fractions

OUT OF 11

Warm-Up · 1/2 | 2/3 | 1/2 | 2/5 | 1/3 | 1/2

Core · 1/6 | 5/12 | 1/3 | 5/12 | 3/8

Challenge · 11/24 | 17/30

2.2 · Borrowing from a Whole

OUT OF 11

Warm-Up · 1/2 | 3/4 | 2/3 | 5/8 | 3/5 | 1/6

Core · 3/2 | 5/4 | 8/3 | 11/8 | 15/4

Challenge · 7/6 | 17/8

2.3 · Mixed Numbers

OUT OF 11

Warm-Up · 3/2 | 5/2 | 5/4 | 1 | 3 | 1

Core · 2 | 5/2 | 2 | 5/2 | 3/4

Challenge · 23/6 | 7/4

2.4 · Fraction Word Problems

OUT OF 11

Warm-Up · 1/2 | 1/2 | 1 | 2/3 | 1/4 | 3/5

Core · 3/4 | 1/4 | 5/6 | 7/6 | 45

Challenge · 5/12 | 40

Fri · Double the Recipe Begins

OUT OF 4

Warm-Up · 1/2 | 1

Core · 4/3 | 3/8

Challenge · 9/8 | 13/12 | 3

Nearest Benchmark

Is it closer to 0, a half, or 1? Decide before computing.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

Is 1/8 nearer 0 or 1	Is 7/8 nearer 0 or 1	4/8 simplified
Is 5/6 nearer 1/2 or 1	Is 2/5 nearer 0 or 1/2	Is 3/5 more or less than 1/2

Core GRADED · SHOW YOUR WORKING

1. Is 5/9 more or less than 1/2
2. Is 4/9 more or less than 1/2
3. Is 7/16 more or less than 1/2
4. Nearest benchmark to 11/12 — type 0, 1/2 or 1
5. Nearest benchmark to 1/9

Challenge NOT GRADED

C1. Is 13/25 more or less than 1/2

C2. Nearest benchmark to 8/15

Estimate the Sum

About a half plus about one is about one and a half.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

Estimate 1/8 + 7/8	Estimate 1/2 + 1/2	Estimate 1/4 + 1/4
Estimate 1/8 + 1/8	1/2 + 1/2	Estimate 7/8 + 7/8

Core GRADED · SHOW YOUR WORKING

1. Estimate 5/6 + 1/8 to the nearest half
2. Estimate 11/12 + 7/8 to the nearest whole
3. Is 2/3 + 1/4 more or less than 1
4. Is 5/6 + 1/3 more or less than 1
5. Estimate 1/9 + 1/10

Challenge NOT GRADED

C1. Is 4/9 + 5/11 more or less than 1

C2. Estimate 7/8 + 1/9 + 1/10 to the nearest whole

Is This Answer Sensible

Given a worked answer, judge it before you check it.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

1/2 + 1/2	1/4 + 1/4	Is 1/2 + 1/3 equal to 2/5 — yes or no
1/2 + 1/3	Is 1/4 + 1/4 equal to 2/8 — yes or no	1/3 + 1/3

Core GRADED · SHOW YOUR WORKING

1. Someone says 2/3 + 3/4 = 5/7. The real answer
2. Someone says 1/2 + 1/4 = 2/6. The real answer
3. Can two fractions under 1 add to more than 1 — yes or no
4. 3/4 + 1/2
5. Is 5/6 + 1/6 equal to 1 — yes or no

Challenge NOT GRADED

C1. Adding numerators and denominators is always wrong — type yes or no

C2. 2/3 + 3/4

Compare Fractions

Common denominator, or reason from benchmarks. Say which.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

Larger: 1/2 or 1/3	Larger: 3/4 or 1/2	Larger: 1/4 or 1/8
Larger: 2/3 or 1/3	Smaller: 1/5 or 1/6	Larger: 5/8 or 3/8

Core

GRADED · SHOW YOUR WORKING

1. Larger: 2/3 or 3/4
2. Larger: 3/5 or 5/8
3. Larger: 5/6 or 7/9
4. Smallest of 1/3, 2/5, 1/4
5. Largest of 2/3, 5/8, 7/12

Challenge

NOT GRADED

C1. Order 3/4, 5/6, 7/8 — type the largest

C2. Larger: 11/15 or 7/10

NAME __________

DATE __________

SCORE ____ / 4

Cook It

Make the scaled recipe for real. Measure honestly.

Warm-Up

2 ITEMS

Double 1/3 cup

Double 1/8 cup

Core

GRADED · SHOW YOUR WORKING

1. Half of 2/3 cup
2. Triple 1/4 cup

Challenge

NOT GRADED

C1. 1 1/2 cups doubled

C2. 2/3 cup plus 3/4 cup

C3. Half of 1 1/2 cups

TEACHER PAGE · ANSWER KEY

Week 3 Answers

Mission 05 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

3.1 · Nearest Benchmark

OUT OF 11

Warm-Up · 0 | 1 | 1/2 | 1 | 1/2 | more

Core · more | less | less | 1 | 0

Challenge · more | 1/2

3.2 · Estimate the Sum

OUT OF 11

Warm-Up · 1 | 1 | 1/2 | 0 | 1 | 2

Core · 1 | 2 | less | more | 0

Challenge · less | 1

3.3 · Is This Answer Sensible

OUT OF 11

Warm-Up · 1 | 1/2 | no | 5/6 | no | 2/3

Core · 17/12 | 3/4 | yes | 5/4 | yes

Challenge · yes | 17/12

3.4 · Compare Fractions

OUT OF 11

Warm-Up · 1/2 | 3/4 | 1/4 | 2/3 | 1/6 | 5/8

Core · 3/4 | 5/8 | 5/6 | 1/4 | 2/3

Challenge · 7/8 | 11/15

Fri · Cook It

OUT OF 4

Warm-Up · 2/3 | 1/4

Core · 1/3 | 3/4

Challenge · 3 | 17/12 | 3/4

Add or Subtract

Mixed set, no signposting. Read before you reach for a method.

NAME __________

DATE __________

SCORE ____ / 11

Warm-Up 6 ITEMS

1/4 + 1/4	3/4 – 1/4	1/3 + 1/3
5/6 – 1/6	1/8 + 3/8	7/8 – 3/8

Core GRADED · SHOW YOUR WORKING

1. 1/2 + 1/3
2. 3/4 – 1/3
3. 2/3 + 1/4
4. 5/6 – 1/2
5. 3/8 + 1/4

Challenge NOT GRADED

C1. 2/3 + 3/4

C2. 5/6 – 3/8

Three Fractions

One common denominator for all three.

NAME __________
DATE __________
SCORE ____ / 11

✦ Warm-Up 6 ITEMS

1/4 + 1/4 + 1/4	1/3 + 1/3 + 1/3	1/8 + 1/8 + 1/8
1/6 + 1/6 + 1/6	1/5 + 1/5 + 1/5	1/2 + 1/4 + 1/4

◆ Core GRADED · SHOW YOUR WORKING

1. 1/2 + 1/3 + 1/6
2. 1/2 + 1/4 + 1/8
3. 1/3 + 1/4 + 1/6
4. 2/3 + 1/6 + 1/6
5. 1/2 + 1/3 + 1/4

★ Challenge NOT GRADED

C1. 1/2 + 1/3 + 1/5

C2. 3/4 + 1/6 + 1/12

Simplify the Answer

6/8 is right. 3/4 is finished.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up
6 ITEMS

2/4 simplified	3/6	4/8
2/6	5/10	3/9

Core
GRADED · SHOW YOUR WORKING

1. 6/8
2. 10/15
3. 12/16
4. 18/24
5. 20/25

Challenge
NOT GRADED

C1. 36/48

C2. 1/4 + 1/4 + 1/4 + 1/4 — simplified

Recipes, Distances, Time

Real contexts, unlike denominators.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

1/2 hour in minutes	1/4 hour in minutes	3/4 hour in minutes
1/2 + 1/4 hour — minutes	1/3 hour in minutes	2/3 hour in minutes

Core

GRADED · SHOW YOUR WORKING

1. Walked 2/3 km then 1/4 km — total
2. A 1/2 hour lesson plus 1/3 hour — minutes
3. 3/4 cup less 1/3 cup
4. Ran 5/8 mile then 1/4 — total
5. 1/6 hour in minutes

Challenge

NOT GRADED

C1. 5/6 hour minus 1/4 hour — minutes

C2. 1/2 + 1/3 + 1/12 hour — minutes

Mid-Unit Quiz

Eight items across Weeks 1–4. 85% to keep flying.

NAME __________

DATE __________

SCORE ____ / 7

✦ Warm-Up
2 ITEMS

1/4 + 1/4

1 – 1/3

◆ Core
GRADED · SHOW YOUR WORKING

1. 1/2 + 1/3
2. 3/4 – 1/3
3. 2/3 = ?/9 — the numerator
4. Is 5/9 more or less than 1/2
5. 12/16 simplified

★ Challenge
NOT GRADED

C1. 2/3 + 3/4

Week 4 Answers

Mission 05 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

4.1 · Add or Subtract

OUT OF 11

Warm-Up · 1/2 | 1/2 | 2/3 | 2/3 | 1/2 | 1/2

Core · 5/6 | 5/12 | 11/12 | 1/3 | 5/8

Challenge · 17/12 | 11/24

4.2 · Three Fractions

OUT OF 11

Warm-Up · 3/4 | 1 | 3/8 | 1/2 | 3/5 | 1

Core · 1 | 7/8 | 3/4 | 1 | 13/12

Challenge · 31/30 | 1

4.3 · Simplify the Answer

OUT OF 11

Warm-Up · 1/2 | 1/2 | 1/2 | 1/3 | 1/2 | 1/3

Core · 3/4 | 2/3 | 3/4 | 3/4 | 4/5

Challenge · 3/4 | 1

4.4 · Recipes, Distances, Time

OUT OF 11

Warm-Up · 30 | 15 | 45 | 45 | 20 | 40

Core · 11/12 | 50 | 5/12 | 7/8 | 10

Challenge · 35 | 55

Fri · Mid-Unit Quiz

OUT OF 7

Warm-Up · 1/2 | 2/3

Core · 5/6 | 5/12 | 6 | more | 3/4

Challenge · 17/12

NAME __________

DATE __________

SCORE ____ / 11

Write Up the Scaling

Every amount, before and after.

Warm-Up
6 ITEMS

Double 1/4	Double 1/3	Double 3/8
Half of 1/2	Half of 2/3	Double 1/8

Core
GRADED · SHOW YOUR WORKING

1. Double 2/3
2. Double 5/6
3. Triple 1/4
4. Half of 3/4
5. Triple 3/8

Challenge
NOT GRADED

C1. Double 1 1/2

C2. Half of 1 1/4

Explain One Conversion

Why that denominator, in one sentence.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up 6 ITEMS

Common denominator of 1/2 and 1/4	Of 1/3 and 1/6	Of 1/2 and 1/3
Of 1/4 and 1/8	Of 1/5 and 1/10	Of 1/2 and 1/5

Core GRADED · SHOW YOUR WORKING

1. Of 2/3 and 3/4
2. Of 5/6 and 3/8
3. Of 3/5 and 1/2
4. Of 7/10 and 2/15
5. Of 1/4, 1/6 and 1/8

Challenge NOT GRADED

C1. Of 2/9 and 5/12

C2. Of 3/7 and 1/2

Mission Review

Everything from five weeks, mixed together.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up

6 ITEMS

1/2 = ?/4 — the numerator	1/4 + 1/4	1 – 1/4
6/8 simplified	Larger: 1/2 or 1/3	5/6 – 1/6

Core

GRADED · SHOW YOUR WORKING

1. 1/2 + 1/3
2. 2/3 + 1/4
3. 3/4 – 1/3
4. 1 1/2 – 3/4
5. Is 4/9 more or less than 1/2

Challenge

NOT GRADED

C1. 1/2 + 1/3 + 1/6

C2. 5/6 – 3/8

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

NAME __________
DATE __________
SCORE ____ / 11

Warm-Up
6 ITEMS

1/3 + 1/3	1 – 1/2	2/4 simplified
1/8 + 3/8	Larger: 3/4 or 1/2	7/8 – 3/8

Core
GRADED · SHOW YOUR WORKING

1. 1/2 + 1/4
2. 2/3 – 1/4
3. 10/15 simplified
4. 1/2 + 1/4 + 1/8
5. Larger: 5/6 or 7/9

Challenge
NOT GRADED

C1. 2/3 + 3/4

C2. 3 – 7/8

Mission 05 Test

Twelve items plus the Big Question, answered out loud.

NAME __________
DATE __________
SCORE ____ / 10

Warm-Up

2 ITEMS

1/4 + 1/4	1 – 1/3

Core

GRADED · SHOW YOUR WORKING

1. 1/2 = ?/6 — the numerator
2. 18/24 simplified
3. 1/2 + 1/3
4. 2/3 + 1/4
5. 3/4 – 1/3
6. 2 – 5/8
7. Larger: 3/5 or 5/8
8. Common denominator of 5/6 and 3/8

Challenge

NOT GRADED

C1. 2/3 + 3/4

C2. 1/2 + 1/3 + 1/4

TEACHER PAGE · ANSWER KEY

Week 5 Answers

Mission 05 · all 5 sets. Answers run in page order. Score out of the graded tiers only — Challenge never counts.

Mark the reasoning, not the handwriting. Any correct method counts.

5.1 · Write Up the Scaling

OUT OF 11

Warm-Up · 1/2 | 2/3 | 3/4 | 1/4 | 1/3 | 1/4

Core · 4/3 | 5/3 | 3/4 | 3/8 | 9/8

Challenge · 3 | 5/8

5.2 · Explain One Conversion

OUT OF 11

Warm-Up · 4 | 6 | 6 | 8 | 10 | 10

Core · 12 | 24 | 10 | 30 | 24

Challenge · 36 | 14

5.3 · Mission Review

OUT OF 11

Warm-Up · 2 | 1/2 | 3/4 | 3/4 | 1/2 | 2/3

Core · 5/6 | 11/12 | 5/12 | 3/4 | less

Challenge · 1 | 11/24

Thu · Error Journal Sweep

OUT OF 11

Warm-Up · 2/3 | 1/2 | 1/2 | 1/2 | 3/4 | 1/2

Core · 3/4 | 5/12 | 2/3 | 7/8 | 5/6

Challenge · 17/12 | 17/8

Fri · Mission 05 Test

OUT OF 10

Warm-Up · 1/2 | 2/3

Core · 3 | 3/4 | 5/6 | 11/12 | 5/12 | 11/8 | 5/8 | 24

Challenge · 17/12 | 13/12